TOYOTA MOTOR NORTH AMERICA

TMNA IT Experiences Process

DATA ENABLEMENT AND GLOBAL COMPLAINCE DATA QUALITY PROCESS

Purpose

Data Enablement and Global Compliance is the group that manages Data Governance at Toyota Motor North America (TMNA). Data Governance is the function to improvise, control, plan, monitor and enforce data management policies. Data Stewards will guide all the data management functions that are performed under the Data Enablement and Global Compliance Program.

The focus is to deploy a disciplined approach to maintain quality of data for better analysis and business decisions. The purpose of the Data Quality Process document is to ensure that high quality data is achieved through guidelines for proper management of enterprise wide information including the Toyota Big Data Platform.

The goal is to deliver a Data Enablement and Global Compliance framework for TMNA to enable trusted, consistent and well-defined information which drives smarter business decisions across the entire information supply chains.

Data Quality is one of the key areas managed by Data Enablement and Global Compliance to ensure superior and efficient data assets. Data Enablement and Global Compliance will be an on-going program to kaizen data assets.

Scope

The Data Quality Process document will describe what is meant by high quality data and will define roles and responsibilities for maintaining, improving and monitoring data quality through an integrated framework of accountability.

The scope of this Data Quality Process document is at the level defined below:

- Define Data Quality Lifecyle, Dimension, classification and threshold
- Define the roles and responsibilities and establish clear lines of accountability
- Develop best practices for effective data management and protection

The primary objective of data quality is to leverage the Data Enablement and Global Compliance Framework for traditional, legacy and big data platforms. The importance of Data Quality is explained below.

- Data, once processed and analyzed, becomes information and information is one of the most valuable resources within any organization as it is used for reporting and decision making.
- In operational terms, “misinformation” could result in inaccurate information and poor business performance. It is essential that the data is of high quality to trust in accurate reporting, decision-making and improvements.
- If the data is of poor quality, and not audit compliant, it could result in

loss of income due to financial penalties.

- Maintaining high quality data is an on-going process with regular assessments that sets national standard for information governance, including data quality and audit compliance

Compliance Compliance with this Process is mandatory. When not meeting this Process's requirements for business reasons, a TMNA manager must seek and obtain an approved Exception, requested and documented through the Exception process in *Toyota's Ticketing Systems*. Team Members who violate this Process may be subject to corrective action, up to and including termination. You are subject to all agreements you signed with the Company, as well as all Company rules and policies, including but not limited to the TMNA Policies.

If you see or hear of an activity that may violate this Process, whether by a Team Member, Corporate Partner, or Third-Party; you must report that activity. If this is a security incident you can email Security@toyota.com or call the Help Desk at (888) 488-4357. Reports may be made anonymously via the Toyota Speak Up Line by calling (844)SPEAK-99 or by visiting the Toyota EthicsPoint website.

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This Process contains the following categories:

1. Policies

Applications not part of Data Enablement and Global Compliance Program.

Item	Requirement	Ref/Flag
1	Data Quality Process will not apply for applications that are not part of Data Enablement and Global Compliance Program	

Applications part of Data Enablement and Global Compliance Program

Item	Requirement	Ref/Flag
1	Applications onboarding on Toyota Big Data Platform (TBDP) as well as traditional and legacy data sources will be part of Data Quality monitoring	
2	Data quality team will follow reactive approach for projecting data anomalies and cleansing it however will follow proactive approach wherever possible	
3	Data quality rules will be defined only on critical data elements (CDE). Data Quality rules will not be defined on every attribute of an application	
4	Every application in data quality scope should be analyzed, profiled and the scores should be published on data quality dashboards on Accuracy, Completeness, Conformity, Uniqueness, Consistency and Timeliness dimensions.	
5	All the Data quality rules defined should be reviewed and approved with the business clients to make sure that they understand them, and that the rules correspond to their business requirements	
6	All the data quality rules defined should be reviewed and approved by Data Stewards	
7	Classified application data (e.g. revenue) should not be displayed in the data quality dashboards	
8	Each data anomaly identified as a part of data quality analysis should be approved and reviewed by data owners to determine if it represents a true data flaw and evaluate the potential business impacts	
9	Establish an automated or event driven process for aligning or possibly correcting flawed data on source or application within business expectations. Data should be cleansed of anomalies identified on traditional or legacy applications. For e.g. VDW, CDW, etc	
10	Access to the data quality dashboards and scorecards should be controlled through active directory groups and all active directory groups created for data quality scorecards should be reviewed and approved by Data Stewards	
11	Changes to the Process shall be reviewed and approved by Data Stewards and Data Quality Owner.	

Item	Requirement	Ref/Flag
12	Communication of Process Changes will be defined by TMNA processes for Team Members and TMNA's Suppliers.	

2. Data Quality Process Exceptions

Item	Requirement	Ref/Flag
1	**Data Quality Threshold** – The threshold levels can be adjusted for a specific data asset of an application and will need application or data owners' approval. For example, a dealer application has a Comments field which can be blank for 50% of time and is still acceptable. In that case the threshold percentage will be reduced to 50% for comments field and will be projected as green instead of red.	
2	**Data Quality Dimension** – It is not mandatory to have all the six data dimensions defined for each application, there can be exceptions for an application which can opt for having data quality rules defined for less than six dimensions. For example, a dealer application needs only consistency and uniqueness dimension. In this case data quality dashboard, will project anomalies only for two dimensions' consistency and uniqueness.	
3	**Access to sensitive data** – Data classified as confidential, highly confidential or legal will be protected and restricted for access to users however exceptions will be granted, and only privileged users will be able to access all classified data.	

3. Roles and Responsibilities

Role	Responsibility
Data Steward	• Data Stewards are responsible for promoting and fostering a culture of data quality. • Ensuring data quality assessments are carried out for each data collection under their area of responsibility. • Reviewing and approving data quality rules to be applied
Data Quality Lead	▪ Interacting with stakeholders' teams and understanding the data quality issues ▪ Detailed requirements gathering and design definition for the data quality framework ▪ Provide technical guidance for performing complex profiling jobs

Role	Responsibility
	▪ Articulate and validate at a detailed level, the proposed solution approach, and perform feasibility analysis and fitment of the proposed solutions accelerators ▪ Ensure data quality thresholds are defined and met, drive plans and remediation activities where thresholds are not met ▪ Provide data quality metrics and updates to management via the use of standardized dashboards
Requestor	▪ Analyze and provide DQ requirements for onboarding application ▪ Provide list of critical data elements (CDE) ▪ Identify and provide applicable DQ rules ▪ Review and validate DQ rules ▪ Analyze and monitor DQ dashboard/scorecards ▪ Prioritize and help resolve anomalies identified as part of DQ reporting
DQ Analyst	▪ Interacting with line of business (LOB) users to gather Application Know-how ▪ Collect information on Current and Target Data Usage by Business ▪ Identify Key Data Elements & user consumption patterns ▪ Perform and elaborate data profiling of the impacted data elements ▪ Analyze reports and share findings ▪ Develop DQ metrics and thresholds ▪ Participate in the data certification process

Role	Responsibility
DQ Developer	▪ Design complex quality rules ▪ Perform scripting for parsing, cleansing, standardization, validation, creating scorecards, exceptions, notifications and reporting standardization ▪ Perform necessary unit testing and participate in deploying scripts, UAT and PROD environment as needed ▪ Resolve the support issues or perform the support tasks on different environments ▪ Monitor dashboards and highlight findings

4. References, Regulations, and Guiding Standards

4.1. Data Quality Lifecycle

Item	Data Quality Lifecycle	Description
1	Discover	• Profile and analyze the data • Identify data content and anomalies.
2	Define	• Define Data Quality Rules • Establish Matrices on Key Performance Indicator's (KPI)
3	Build	• Design and implement data quality rules and Processes. • Establish exception handling framework.
4	Remediate	• Review Exceptions and Refine rules. • Remediate and fix the data.
5	Monitor	• Monitor Data Quality Progress (DQ scorecard, Reports and Dash Boards).

4.2. Data Quality Dimensions

Item	Data Quality Dimension	Description
1	Completeness	To check if the requisite information is available. E.g. – Percentage NULL present in Birthdate column. If the column contains 50% NULL, the data is not complete.

Item	Data Quality Dimension	Description
2	Accuracy	To check if data objects accurately represent the “real-world” values they are expected to model. E.g. In the Birth Date column year is missing “08-11” and expected value is DD-MM-YYYY then the birthdate column is not accurate.
3	Conformity	To check if the Data presented is in specific format. E.g. Birthdate is expected in YYYY-MM-DD format and the value is “08-12-1985” (MM-DD-YYYY). The data fails on the conformity dimension.
4	Consistency	To check if data is same across different underlying source systems. E.g.: Source A, Source B and TBDP (Toyota Big Data Platform) should have exactly same data. If birthdate column in Source A, B and TBDP are inconsistent then the data is not consistent. Source A – 08-12 Source B – 08-12-1985 TBDP – 12-08
5	Uniqueness	To check if there is any duplicate data reported. E.g.: In a Table there are two columns. Field A is primary key and should not have any duplicates however there are more than one records for COL_A, in that case the attribute fails on uniqueness dimension. COL_A Total COUNT ABC 5 XYX 3
6	Timeliness	To check the difference between data capture across underlying source/application system. E.g.: The date modified for Source A and TBDP have a difference of 1 year then this field fails on timeliness dimension. The date modified should be same for source A and TBDP. In below case, the date modified is same however the values are different for COL_B so this field fails in timeliness dimension. Source A: COL_A COL_B DATE_MODIFIED 45 20 2017-01-01 TBDP:

Item	Data Quality Dimension	Description
		COL_A COL_B DATE_MODIFIED 45 22 2017-01-01

4.3. Data Quality Thresholds

Item	Threshold	Display Color	Description
1	95-100%	Green	The data is Good. For e.g.: In completeness dimension, Name field has 95% valid data and only 5% of data is NULL which is acceptable.
2	85-94%	Yellow	This data is Acceptable. For e.g.: In conformity dimension, birthdate field, only 10-15% of data values are not in YYYY-DD-format, but 85% data is in the given format so that data is good.
3	0-84%	Red	This data is unacceptable. For e.g.: In accuracy dimension, birthdate field has more than 15% of invalid data that means the quality of field is not good.

4.4. Active Directory Groups

Item	Groups	Permission		
		DEV	**QA**	**PROD**
1	Consumer/User	Read	Read	Read
2	Data Admins	Read Write Execute	Read Write Execute	Read Write Execute
3	Data Analyst	Read Execute	Read Execute	Read Execute
4	Data Architects	Read	Read	Read
5	Data Owners/Business Owners	Read	Read	Read
6	Data Stewards	Read Execute	Read Execute	Read Execute
7	Developer	Read Write Execute	Read Execute	Read Execute

5. Revision History

Version	Date	Author	Description	Section(s) Affected
1.0	08/30/2017	DataEnablement@toyota.com	The Data Quality Process document is to deploy a disciplined approach to maintain quality of data for better analysis and business decisions.	All
1.0	10/15/2021	DataEnablement@toyota.com	Added Process in new ITx Process template	All

6. Appendix for Relevant Processes

Data Enablement ITx Metadata Process

[TMNA_DE_Metadata_Process.pdf (toyota.com)]